



Week 3 : Task 8

Network Scanning With Nmap

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Date of Submission

04/07/2026

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Tool Used: Nmap (Network Mapper)

Operating System: Kali Linux

Target Machine: Metasploitable 2

Target IP Address: 192.168.1.6

Objective

The objective of this task was to learn the fundamentals of network reconnaissance using Nmap. The task involved discovering live hosts, identifying open ports and running services, understanding aggressive scanning techniques, and documenting the scan results in a structured format.

Introduction

Nmap (Network Mapper) is one of the most widely used open-source network scanning tools. Security professionals and penetration testers use Nmap to discover devices connected to a network, identify open ports, detect running services, determine operating systems, and perform security assessments.

During this practical exercise, Kali Linux was used to scan the Metasploitable virtual machine in a controlled lab environment.

Lab Environment

Component	Details
Attacker Machine	Kali Linux
Target Machine	Metasploitable 2
Virtualization	Oracle VirtualBox
Scanner	Nmap 7.98
Target IP	192.168.1.6

Task 1 – Host Discovery

Objective

Identify all active hosts available within the local network.

Command Used

```
nmap -sn 192.168.1.0/24
```

Explanation

The `-sn` option performs a Ping Scan without scanning ports. It checks which systems are alive on the specified subnet.

The `/24` CIDR notation represents the network range from **192.168.1.1** to **192.168.1.254**.

This scan helps penetration testers quickly identify active devices before beginning detailed enumeration.

```
rtt min/avg/max/mdev = 0.358/0.665/1.281/0.267 ms

(hamza@hamza)-[/usr/share/nmap/scripts]
$ nmap -sn 192.168.1.0/24
Starting Nmap 7.98 ( https://nmap.org ) at 2026-07-17 14:25 -0400
Nmap scan report for 192.168.1.1
Host is up (0.0026s latency).
MAC Address: 52:54:00:12:35:00 (QEMU virtual NIC)
Nmap scan report for 192.168.1.2
Host is up (0.00035s latency).
MAC Address: 08:00:27:7E:C1:0A (Oracle VirtualBox virtual NIC)
Nmap scan report for 192.168.1.6
Host is up (0.00060s latency).
MAC Address: 08:00:27:09:CA:FC (Oracle VirtualBox virtual NIC)
Nmap scan report for 192.168.1.3
Host is up.
Nmap done: 256 IP addresses (4 hosts up) scanned in 7.63 seconds

(hamza@hamza)-[/usr/share/nmap/scripts]
$ nmap 192.168.1.6
Starting Nmap 7.98 ( https://nmap.org ) at 2026-07-17 14:26 -0400
Nmap scan report for 192.168.1.6
Host is up (0.00026s latency).
Not shown: 977 closed tcp ports (reset)
PORT      STATE SERVICE
21/tcp    open  ftp
22/tcp    open  ssh
23/tcp    open  telnet
25/tcp    open  smtp
53/tcp    open  domain
80/tcp    open  http
111/tcp   open  rpcbind
139/tcp   open  netbios-ssn
445/tcp   open  microsoft-ds
512/tcp   open  exec
513/tcp   open  login
514/tcp   open  shell
1099/tcp  open  rmiregistry
1524/tcp  open  ingreslock
2049/tcp  open  nfs
2121/tcp  open  ccproxy-ftp
3306/tcp  open  mysql
5432/tcp  open  postgresql
5900/tcp  open  vnc
6000/tcp  open  x11
```

Observation

The scan successfully detected the active hosts connected to the local network. One of the discovered systems was the Metasploitable virtual machine with the IP address **192.168.1.6**.

Task 2 – Service Version Detection

Objective

Identify open ports and determine the services and software versions running on the target system.

Command Used

```
nmap -sV 192.168.1.6
```

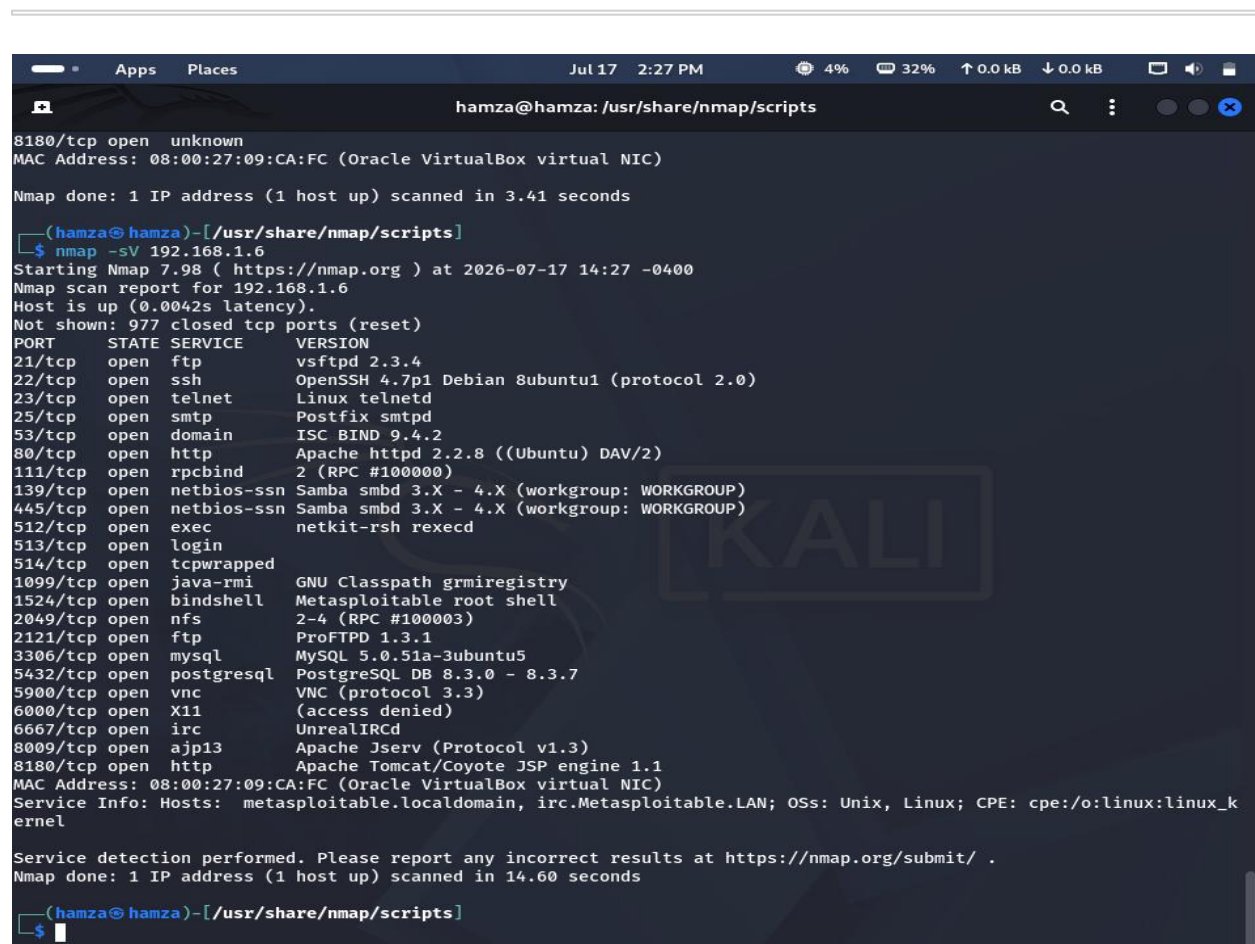
Explanation

The `-sV` option enables Service Version Detection.

Nmap connects to every open port and attempts to identify:

- Running service
- Software name
- Software version

Knowing service versions allows security professionals to identify outdated software and known vulnerabilities.



```
hamza@hamza: /usr/share/nmap/scripts
8180/tcp open unknown
MAC Address: 08:00:27:09:CA:FC (Oracle VirtualBox virtual NIC)
Nmap done: 1 IP address (1 host up) scanned in 3.41 seconds

(hamza@hamza) - [/usr/share/nmap/scripts]
$ nmap -sV 192.168.1.6
Starting Nmap 7.98 ( https://nmap.org ) at 2026-07-17 14:27 -0400
Nmap scan report for 192.168.1.6
Host is up (0.0042s latency).
Not shown: 977 closed tcp ports (reset)
PORT      STATE SERVICE        VERSION
21/tcp    open  ftp            vsftpd 2.3.4
22/tcp    open  ssh            OpenSSH 4.7p1 Debian 8ubuntu1 (protocol 2.0)
23/tcp    open  telnet         Linux telnetd
25/tcp    open  smtp           Postfix smtpd
53/tcp    open  domain         ISC BIND 9.4.2
80/tcp    open  http           Apache httpd 2.2.8 ((Ubuntu) DAV/2)
111/tcp   open  rpcbind        2 (RPC #100000)
139/tcp   open  netbios-ssn    Samba smbd 3.X - 4.X (workgroup: WORKGROUP)
445/tcp   open  netbios-ssn    Samba smbd 3.X - 4.X (workgroup: WORKGROUP)
512/tcp   open  exec           netkit-rsh rexecd
513/tcp   open  login          netkit-rsh rlogind
514/tcp   open  tcpwrapped
1099/tcp  open  java-rmi       GNU Classpath grmiregistry
1524/tcp  open  bindshell      Metasploitable root shell
2049/tcp  open  nfs            2-4 (RPC #100003)
2121/tcp  open  ftp            ProFTPD 1.3.1
3306/tcp  open  mysql          MySQL 5.0.51a-3ubuntu5
5432/tcp  open  postgresql     PostgreSQL DB 8.3.0 - 8.3.7
5900/tcp  open  vnc            VNC (protocol 3.3)
6000/tcp  open  X11            (access denied)
6667/tcp  open  irc            UnrealIRCd
8009/tcp  open  ajp13          Apache Jserv (Protocol v1.3)
8180/tcp  open  http           Apache Tomcat/Coyote JSP engine 1.1
MAC Address: 08:00:27:09:CA:FC (Oracle VirtualBox virtual NIC)
Service Info: Hosts: metasploitable.localdomain, irc.Metasploitable.LAN; OSs: Unix, Linux; CPE: cpe:/o:linux:linux_kernel

Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 14.60 seconds

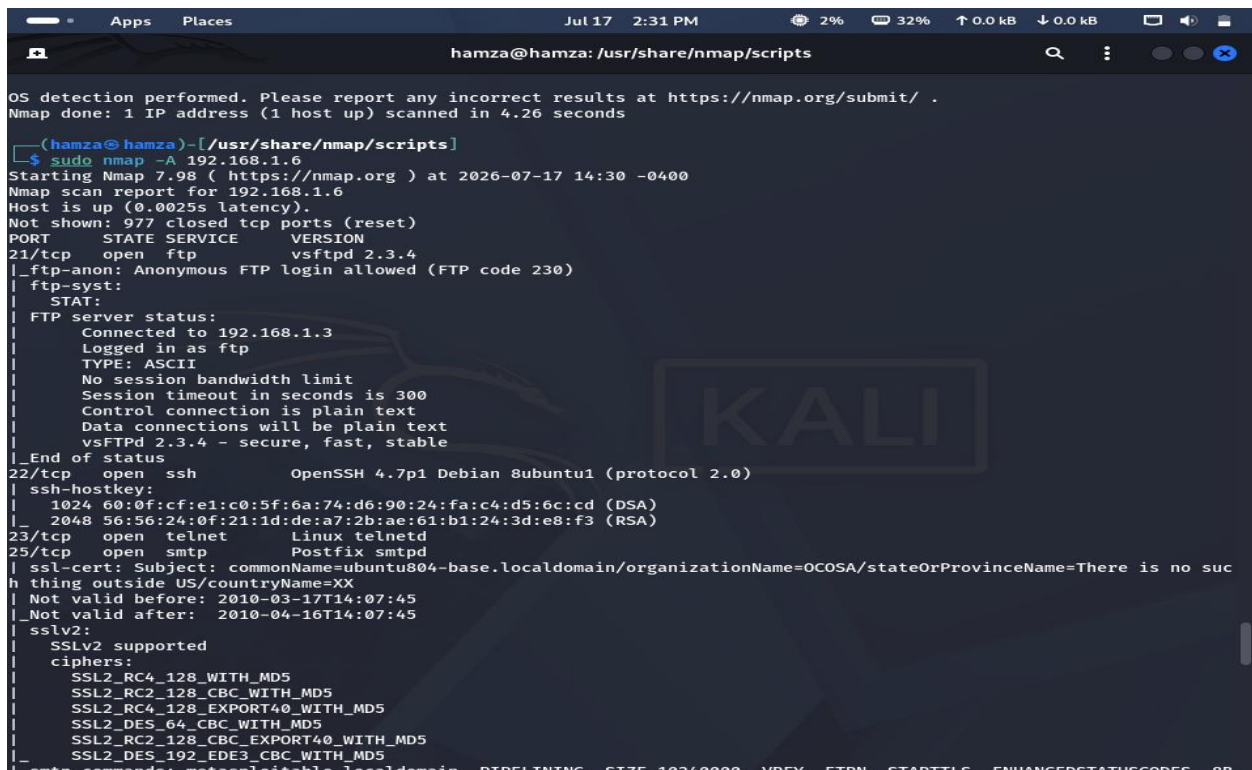
(hamza@hamza) - [/usr/share/nmap/scripts]
$
```

Sample Scan Results

Port	State	Service	Version
21	Open	FTP	vsftpd
22	Open	SSH	OpenSSH
23	Open	Telnet	Linux Telnet
25	Open	SMTP	Postfix
53	Open	DNS	ISC BIND
80	Open	HTTP	Apache HTTP Server
111	Open	RPCbind	rpcbind
139	Open	NetBIOS	Samba
445	Open	SMB	Samba
3306	Open	MySQL	MySQL Database
5432	Open	PostgreSQL	PostgreSQL Database
5900	Open	VNC	VNC Server
6000	Open	X11	X Window System
6667	Open	IRC	UnrealIRCd
8009	Open	AJP13	Apache JServ
8180	Open	HTTP	Apache Tomcat

Observation

The scan identified multiple open TCP ports exposing different network services. Several of these services are intentionally vulnerable in Metasploitable and are commonly used for penetration testing practice.



Information Collected

The Aggressive Scan provided:

- Host Status
 - Open TCP Ports
 - Running Services
 - Service Versions
 - Operating System Detection
 - Traceroute Information
 - NSE Script Results
-

Observation

The Aggressive Scan revealed detailed information about the Metasploitable machine. Nmap successfully identified the Linux operating system, detected service versions, and executed default NSE scripts to gather additional information.

Open Ports Summary

Port	Protocol	State	Service	Description
21	TCP	Open	FTP	File Transfer Protocol
22	TCP	Open	SSH	Secure Shell
23	TCP	Open	Telnet	Remote Login
25	TCP	Open	SMTP	Email Service
53	TCP	Open	DNS	Domain Name System
80	TCP	Open	HTTP	Web Server
111	TCP	Open	RPCbind	Remote Procedure Calls
139	TCP	Open	NetBIOS	Windows File Sharing
445	TCP	Open	SMB	Server Message Block
3306	TCP	Open	MySQL	Database Service

Port	Protocol	State	Service	Description
5432	TCP	Open	PostgreSQL	Database Service
5900	TCP	Open	VNC	Remote Desktop
6000	TCP	Open	X11	Linux GUI
6667	TCP	Open	IRC	Internet Relay Chat
8009	TCP	Open	AJP13	Apache JServ Protocol
8180	TCP	Open	HTTP	Apache Tomcat

Tools Used

- Kali Linux
- Oracle VirtualBox
- Nmap
- Metasploitable 2

Resources

- TryHackMe — Nmap (Free Room)
- Nmap Official Documentation (`man nmap`)

YouTube References

- Nmap Tutorial for Beginners: Nmap Basics
- Nmap Scanning Explained for Beginners

Learning Outcomes

After completing this task, I learned how to:

- Discover active hosts on a network using Ping Scan.
- Detect open ports and running services.

- Identify service versions using Nmap.
 - Perform an Aggressive Scan to gather detailed target information.
 - Interpret Nmap scan results.
 - Create a professional summary table of detected services.
 - Understand the importance of network reconnaissance in penetration testing.
-

Conclusion

This practical exercise demonstrated the importance of Nmap in the reconnaissance phase of ethical hacking. Using Kali Linux, I successfully scanned the Metasploitable virtual machine to discover live hosts, identify open ports, detect running services and software versions, and collect detailed information through Aggressive Scan mode. The results highlighted how Nmap provides valuable insights into a target system, enabling security professionals to understand the attack surface and plan further security assessments in an authorized testing environment.

Resources and Learning Materials

Objective

The purpose of using these resources was to understand the fundamentals of Nmap, learn different scanning techniques, and perform practical network reconnaissance in a controlled lab environment.

Resource 1: TryHackMe – Nmap (Free Room)

Description

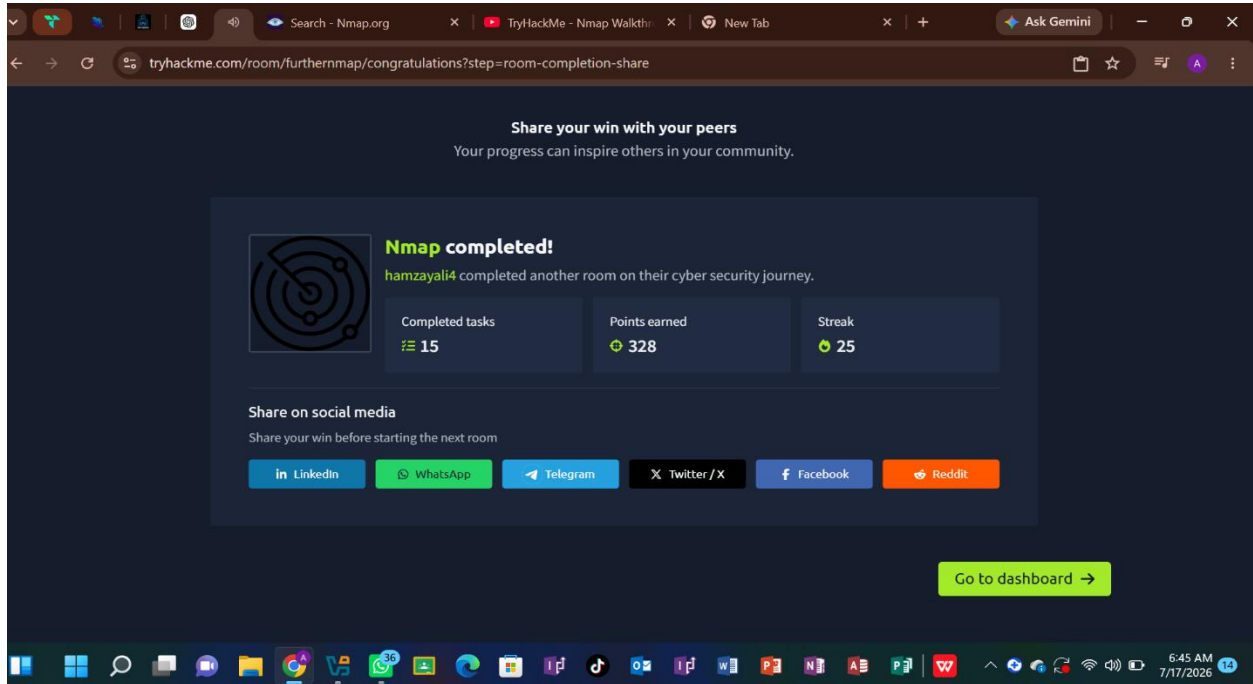
The TryHackMe Nmap room provides a practical introduction to Nmap. It explains how Nmap is used for host discovery, port scanning, service detection, operating system identification, and network reconnaissance. The room combines theoretical concepts with hands-on exercises, making it easier to understand how different Nmap scan types work.

Topics Covered

- Introduction to Nmap
- Host Discovery
- TCP and UDP Port Scanning
- Service Version Detection
- Operating System Detection
- NSE (Nmap Scripting Engine)
- Aggressive Scanning
- Scan Timing and Performance

Learning Outcome

After completing the TryHackMe room, I learned how to discover active hosts, identify open ports, detect running services, and understand how Nmap assists penetration testers during the reconnaissance phase of a security assessment.



Resource 2: Nmap Official Reference (Man Page)

Description

The official Nmap manual (`man nmap`) provides complete documentation for all Nmap commands, switches, and scanning techniques. It is one of the most reliable references for understanding the functionality of each option available in Nmap.

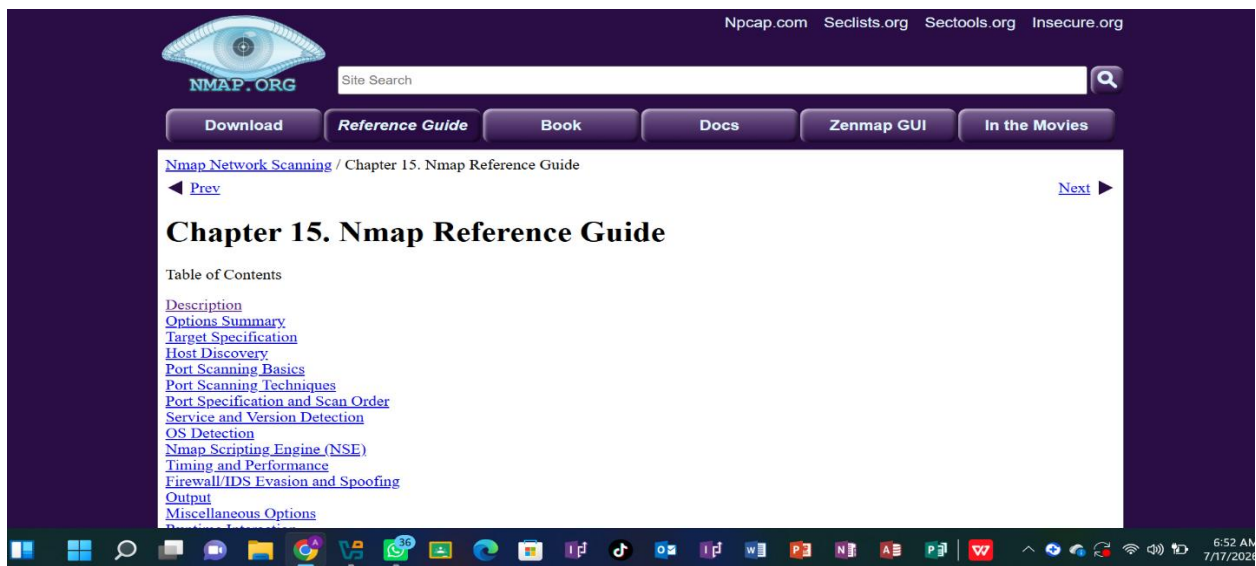
Frequently Used Commands

Command	Purpose
<code>nmap -sn</code>	Discover live hosts
<code>nmap -sV</code>	Detect service versions
<code>nmap -O</code>	Detect operating system

Command	Purpose
<code>nmap -A</code>	Perform aggressive scan
<code>nmap -p-</code>	Scan all TCP ports
<code>nmap --script vuln</code>	Run vulnerability detection scripts

Learning Outcome

The official documentation helped in understanding the syntax, command options, and best practices for performing different types of network scans.



Practical Learning

Host Discovery

The following command was used to identify active hosts within the local network.

```
nmap -sn 192.168.1.0/24
```

Screenshot

Figure 1: Host Discovery Using Nmap (-sn)

```
hamza@hamza: /usr/share/nmap/scripts
rtt min/avg/max/mdev = 0.358/0.665/1.281/0.267 ms

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Host is up (0.00035s latency).
MAC Address: 08:00:27:7E:C1:0A (Oracle VirtualBox virtual NIC)
Nmap scan report for 192.168.1.6
Host is up (0.00060s latency).
MAC Address: 08:00:27:09:CA:FC (Oracle VirtualBox virtual NIC)
Nmap scan report for 192.168.1.3
Host is up.
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Host is up (0.00026s latency).
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PORT      STATE SERVICE
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22/tcp    open  ssh
23/tcp    open  telnet
25/tcp    open  smtp
53/tcp    open  domain
80/tcp    open  http
111/tcp   open  rpcbind
139/tcp   open  netbios-ssn
445/tcp   open  microsoft-ds
512/tcp   open  exec
513/tcp   open  login
514/tcp   open  shell
1099/tcp  open  rmiregistry
1524/tcp  open  ingreslock
2049/tcp  open  nfs
2121/tcp  open  ccproxy-ftp
3306/tcp  open  mysql
5432/tcp  open  postgresql
5900/tcp  open  vnc
6000/tcp  open  x11
```

Observation

The scan successfully identified the active devices connected to the local network. This technique is useful during the initial reconnaissance phase because it quickly determines which hosts are online without performing a port scan.

Video Learning

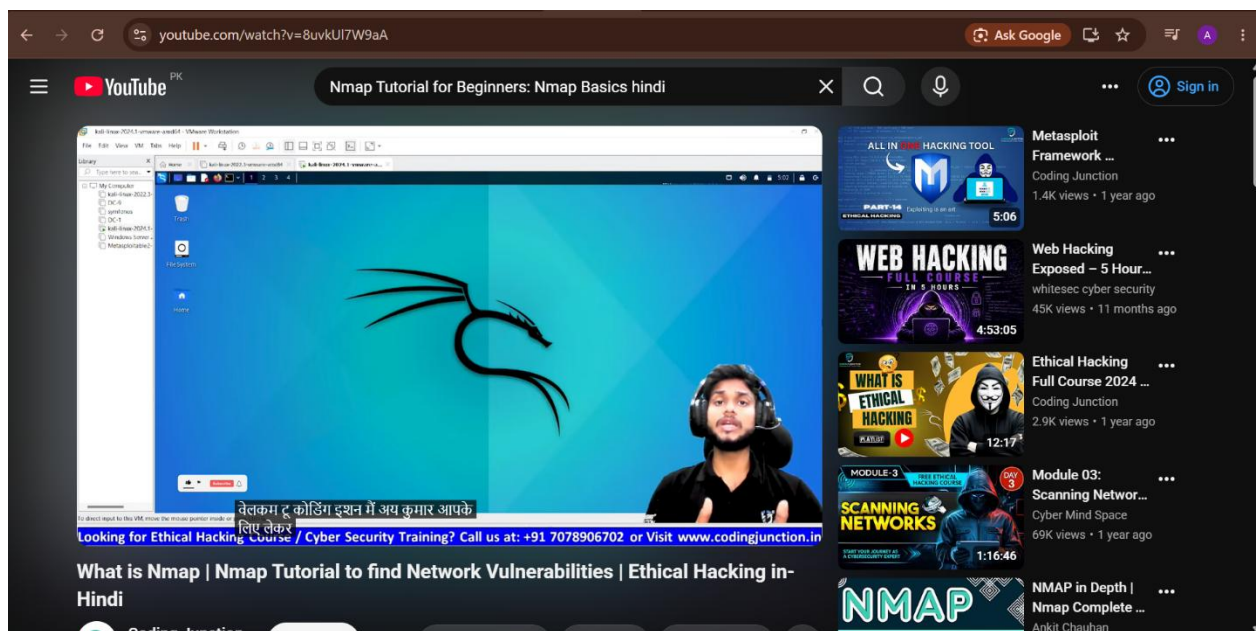
[Video 1: Nmap Tutorial for Beginners: Nmap Basics](#)

Summary

This tutorial explains the basic concepts of Nmap, including host discovery, TCP scanning, service detection, and common command-line options. It demonstrates how Nmap is used to gather information about a target system during the reconnaissance phase.

Key Concepts Learned

- Basic Nmap syntax
- Network host discovery
- TCP port scanning
- Understanding scan results
- Reading Nmap output



Video 2: Nmap Scanning Explained for Beginners

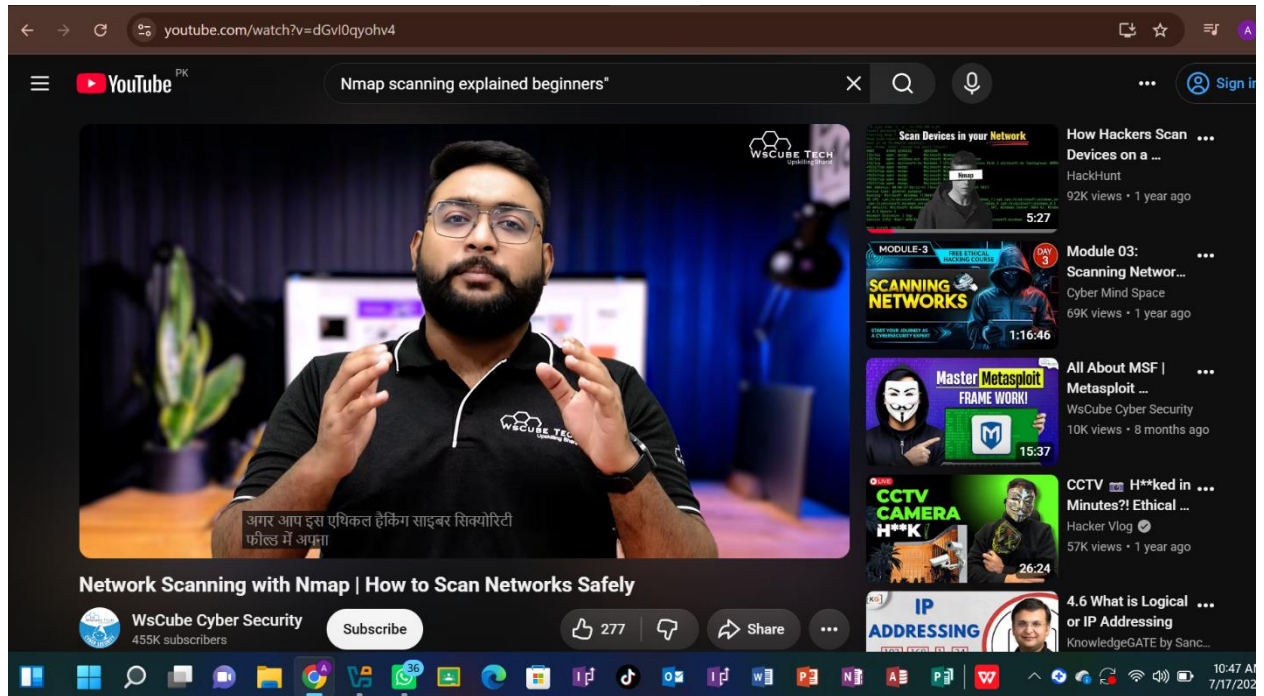
Summary

This video provides a detailed explanation of different scanning techniques such as SYN Scan, TCP Connect Scan, UDP Scan, and Aggressive Scan. It also explains how Nmap communicates with target systems and interprets network responses.

Key Concepts Learned

- SYN Scan (`-sS`)
- TCP Connect Scan (`-sT`)
- UDP Scan (`-sU`)
- Version Detection (`-sV`)
- Aggressive Scan (`-A`)

- Nmap Scripting Engine (NSE)



Overall Learning

By studying the TryHackMe room, reading the official Nmap documentation, and watching beginner tutorials, I developed a solid understanding of network reconnaissance using Nmap. I learned how to discover live hosts, identify open ports, detect service versions, interpret scan results, and use Nmap effectively in a controlled penetration testing environment. These resources provided both theoretical knowledge and practical skills that are essential for ethical hacking and cybersecurity assessments.